PAL API Guide

This guide introduces PAL through a working model, then explains each component and its behavioral contracts. Start with the component overview and follow the worked tutorial in order. The later sections document detailed behavior, constraints, and specialized operations. For current public signatures and method descriptions, use the API Reference. For quick syntax, use the Cheatsheet. For implementation structure and design rationale, see PAL Architecture and Design.

Choose the right PAL structure

Need	PAL structure
One typed value per lattice site	`Grid`
Individual agents with identity/properties	`AgentGrid`
Integer counts without individual identity	`PopGrid`
Continuous diffusing/advecting field	`PDEgrid`
Reusable integer query result	`IList`
Binomial/multinomial draws	`Multinomial`

These components can work together in one model. For example, individual cells in an `AgentGrid` can consume a nutrient represented by a `PDEgrid`. Some components support accumulating changes before applying them together with `Update()`; later sections explain why the timing of those updates matters.

Build a cell model, one feature at a time

This tutorial follows **one simple model**: cells occupy sites on a two-dimensional lattice. First we place and draw them. Then we let them move to empty neighboring sites. Then we compile the movement rule, add a boundary-fed nutrient field, replace agents with population counts, and save the state. Each stage reuses the same biological idea, so the new material is about PAL rather than new biology.

Use the examples in order. The code blocks marked **complete program** can be saved as separate Python files and run independently. Later snippets show focused modifications to the preceding program. The detailed API sections remain available for complete behavior and less common features.

PAL begins in safe mode, which checks operations for common errors. The tutorial uses this default. Later, the performance section explains how to select fast mode and compile frequently executed code.

1. Place cells on a lattice

An `AgentGrid` stores identifiable cells and their positions. We choose a nonstackable 30 × 30 lattice: at most one cell can occupy each site. A `Grid`, by contrast, stores one typed value per site and does not create individual cell identities.

Complete program — initial cells:

```
import PythonAutomataLibrary as pal

SIZE = 30

def Setup():
    cells = pal.NewAgentGrid((SIZE, SIZE), isStackable=False)
    for x, y in ((14, 14), (15, 14), (14, 15), (15, 15)):
        cells.NewAgentSQ(x, y)
    return cells

def main():
    cells = Setup()
    print("Initial cells:", len(cells.All()))

if __name__ == "__main__":
    main()
```

`NewAgentSQ(x, y)` places a cell at lattice coordinates `(x, y)`. An agent has an identity (its handle) as well as a position. The lattice dimensions are not the number of cells: most sites are initially empty.

2. Draw the cells immediately

Visual feedback is useful from the first model. `StartPixWindow` gives us a pixel grid and a window. `pal.AwaitWindows()` waits for the windows to be ready before the program continues. Each site becomes one pixel, enlarged by `scale`. We draw black background pixels and red occupied sites.

Complete program — place and display cells:

```
import time
import PythonAutomataLibrary as pal

SIZE = 30
BLACK = pal.RGB(0, 0, 0)
RED = pal.RGB(255, 0, 0)

def Draw(cells, pix):
    pix[:] = BLACK
    for agent in cells.All():
        pix[cells.XSQ(agent), cells.YSQ(agent)] = RED

def Setup():
    cells = pal.NewAgentGrid((SIZE, SIZE), isStackable=False)
    for x, y in ((14, 14), (15, 14), (14, 15), (15, 15)):
        cells.NewAgentSQ(x, y)
    return cells

def main():
    cells = Setup()
    pix, window = pal.StartPixWindow(SIZE, SIZE, scale=12, title="Cells")
    Draw(cells, pix)
    window.Update()
    pal.AwaitWindows()
    while window.IsOpen():
        time.sleep(0.05)
    window.Close()
```

```
if __name__ == "__main__":
    main()
```

You should see four red cells near the center. If you change their initial coordinates, the drawing changes accordingly. `window.Update()` publishes the pixels you wrote; drawing into `pix` alone does not refresh the display. Later we will call `Draw` after every timestep.

3. Find neighbors and empty sites

In this model, cells move only into empty neighboring sites. `pal.MooreHood(2, excludeCenter=True)` describes the eight relative offsets around a 2D site. `cells.Hood(..., x, y)` converts those offsets to valid lattice coordinates. `XSQ` and `YSQ` give an agent's lattice site.

An `IList` is PAL's reusable list of integers. We use it to collect empty neighboring sites. Each site has coordinates `(x, y)`, but `IList` stores a single integer per site: `cells.ToI(x, y)` converts coordinates to that site's index. Later, `cells.ItoX(i)` and `cells.ItoY(i)` convert the index back to coordinates. Add the following code to the program from stage 2, immediately after the color constants:

```
MOORE = pal.MooreHood(2, excludeCenter=True)

def EmptyNeighbors(cells, agent, empty):
    empty.Clear()
    for x, y in cells.Hood(MOORE, cells.XSQ(agent), cells.YSQ(agent)):
        if cells.counts[x, y] == 0:
            empty.Append(cells.ToI(x, y))
    return empty
```

Create `empty = pal.NewIList()` once in `main`, immediately after `cells = Setup()`. To inspect the result before adding movement, insert `print(EmptyNeighbors(cells, cells.All()[0], empty).All())` immediately afterward. The printed indices identify available lattice sites; `ItoX` and `ItoY` recover their coordinates. At a nonperiodic boundary, `Hood` omits offsets outside the grid.

4. Move cells to empty neighboring sites

Each cell takes a random walk: at every timestep, it chooses one available empty neighboring site and moves there. No cells are created or removed. We iterate a shuffled snapshot of existing agents; moves update occupancy immediately, so later cells see the new positions.

Continue with the program modified in stage 3, retaining `MOORE`, `EmptyNeighbors`, and the reusable `empty` list. Remove the optional diagnostic `print` if you added it. Define the following `Step` function alongside `Draw` and `Setup`:

```
def Step(cells, empty):
    for agent in cells.All(shuffle=True):
        EmptyNeighbors(cells, agent, empty)
        if len(empty):
            site = empty.Random()
            cells.MoveSQ(agent, cells.ItoX(site), cells.ItoY(site))
```

Replace the stationary display loop with:

```
    pal.AwaitWindows()
    while window.IsOpen():
        Step(cells, empty)
        Draw(cells, pix)
        window.Update()
        time.sleep(0.05)
    window.Close()
```

The cells wander without overlapping. A cell with no empty neighbors stays put. Selecting uniformly among *available* sites means the probability of each direction depends on occupancy.

5. Compile the timestep

Because the neighborhood results already use `IList`, compiling the movement rule requires no change in data representation. Add `@pal.njit(cache=True)` to the same `Step` function and annotate its PAL arguments:

```
@pal.njit(cache=True)
def Step(cells: pal.AgentGrid, empty: pal.IList):
    for agent in cells.All(shuffle=True):
        EmptyNeighbors(cells, agent, empty)
        if len(empty):
            site = empty.Random()
            cells.MoveSQ(agent, cells.ItoX(site), cells.ItoY(site))
```

Replace the earlier `EmptyNeighbors` definition with this compiled version as well:

```
@pal.njit(cache=True)
def EmptyNeighbors(cells: pal.AgentGrid, agent: int, empty: pal.IList):
    empty.Clear()
    for x, y in cells.Hood(MOORE, cells.XSQ(agent), cells.YSQ(agent)):
        if cells.counts[x, y] == 0:
            empty.Append(cells.ToI(x, y))
    return empty
```

Both functions now run in compiled code, while `main`, window creation, and rendering can remain in ordinary Python. The simulation rule and data representation are unchanged. `cache=True` can reuse compiled specializations in later runs; the performance section explains the tradeoffs.

6. Add a diffusing nutrient

The cells now share a continuous nutrient field, represented by a `PDEgrid` on the same lattice. Nutrient **enters from the four boundaries**, diffuses inward, and is **consumed wherever a cell is present**. This behavior will remain unchanged when we replace individual agents with population counts in the next stage.

Add this alongside the cell-grid construction in `main`:

```
nutrient = pal.NewPDEgrid((SIZE, SIZE))
nutrient.SetTimeSpaceStep(1.0, 1.0, 1.0)
nutrient[:] = 1.0
```

The initial concentration is 1 everywhere. Each timestep, cells remove nutrient locally and the boundaries maintain an exterior concentration of 1:

```
def StepNutrient(cells, nutrient):
    for agent in cells.All():
        x, y = cells.XSQ(agent), cells.YSQ(agent)
        nutrient[x, y] = max(0.0, nutrient[x, y] - 0.01)
    nutrient.Diffusion(0.1, xMinBC=1.0, xMaxBC=1.0, yMinBC=1.0, yMaxBC=1.0)
    nutrient.Update()
```

Keep the movement step from stage 5. In `main`, construct `nutrient` once alongside `cells`, then add the nutrient update to the existing simulation loop in this order:

```
Step(cells, empty)                # move cells first
StepNutrient(cells, nutrient)     # consume and diffuse nutrient
Draw(cells, pix)                 # display the updated cells
window.Update()
```

`StepNutrient` can remain an ordinary Python function for this first version; compiling the cell movement step does not require compiling every other function. The boundary arguments specify fixed exterior concentrations: unlike the default no-flux boundaries, these permit nutrient to diffuse into the modeled region when cells deplete it. `Diffusion` queues the transport changes and `Update()` commits them. The values are illustrative, not biologically calibrated.

To see nutrient depletion, open a second `PixWindow` and color each site according to `nutrient[x, y]`, for example using `pal.ColorScale`; the cell window can remain unchanged. See Visualization for rendering and Boundary conditions for transport for boundary options.

7. Replace individual cells with population counts

Now relax the **one cell per site** constraint. With multiple occupancy allowed, cells no longer need to search for empty destinations. A `PopGrid` holds counts instead of individual agents, and a `Multinomial` can distribute each site's cells among the eight directions of the same random walk. There is no division or death: the total number of cells remains constant. The nutrient field from stage 6 is retained, including its boundary supply and cellular consumption.

This stage starts a separate, complete program rather than modifying the stage 6 program in place. Replace the individual-agent grid and its empty-neighbor movement rule with the count-based implementation below. You can keep the stage 6 version separately to compare their behavior.

Complete program — count-based random movement:

```
import time
import PythonAutomataLibrary as pal

SIZE = 30
MOORE = pal.MooreHood(2, excludeCenter=True)
BLACK = pal.RGB(0, 0, 0)
RED = pal.RGB(255, 0, 0)

@pal.njit(cache=True)
def Step(cells: pal.PopGrid, multi: pal.Multinomial):
    for i in cells.All():
```

```
        multi.Setup(cells[i])
        x, y = cells.ItoX(i), cells.ItoY(i)
        for nx, ny in cells.Hood(MOORE, x, y):
            moved = multi.Sample(1.0 / 8)
            cells.Add(-moved, i)
            cells.Add(moved, nx, ny)
    cells.Update()

@pal.njit(cache=True)
def StepNutrient(cells: pal.PopGrid, nutrient: pal.PDEgrid):
    for i in range(len(cells)):
        nutrient[i] = max(0.0, nutrient[i] - 0.01 * cells[i])
    nutrient.Diffusion(0.1, xMinBC=1.0, xMaxBC=1.0, yMinBC=1.0, yMaxBC=1.0)
    nutrient.Update()

@pal.njit(cache=True)
def Draw(cells: pal.PopGrid, pix: pal.Pix):
    pix[:] = BLACK
    for x in range(SIZE):
        for y in range(SIZE):
            if cells[x, y]:
                pix[x, y] = RED

def main():
    pal.Seed(1234)
    cells = pal.NewPopGrid((SIZE, SIZE))
    for x, y in ((14, 14), (15, 14), (14, 15), (15, 15)):
        cells[x, y] = 1
    multi = pal.NewMultinomial()
    nutrient = pal.NewPDEgrid((SIZE, SIZE))
    nutrient.SetTimeSpaceStep(1.0, 1.0, 1.0)
    nutrient[:] = 1.0
    pix, window = pal.StartPixWindow(SIZE, SIZE, scale=12, title="Population random walk")
    pal.AwaitWindows()
    while window.IsOpen():
        Step(cells, multi)
        StepNutrient(cells, nutrient)
        Draw(cells, pix)
        window.Update()
        time.sleep(0.05)
    window.Close()

if __name__ == "__main__":
    main()
```

Each of the eight Moore-neighborhood directions has probability `1/8`. `Setup(cells[i])` starts a multinomial allocation for all cells currently at site `i`; each `Sample(1.0 / 8)` assigns cells to one valid direction. At a nonwrapped boundary, invalid directions are skipped, and their unallocated probability mass corresponds to cells staying at their original site. Probabilities refer to the **original** site's population, not the remaining unassigned cells. Queued transfers are committed together with `Update()`, so arrivals do not move again in the same timestep.

The nutrient update now consumes 0.01 concentration units per cell at each site, so multiple occupancy increases local consumption. The same fixed-value boundaries replenish nutrient in both representations. The two versions implement the same basic random-walk idea, but not the same exclusion process: `AgentGrid` avoids occupied destinations and updates agents sequentially; `PopGrid` permits multiple occupancy and moves all counts synchronously. That distinction matters when comparing results. See Multinomial for more about sampling and solver reuse.

8. Reproduce and save a run

For reproducible runs, call `pal.Seed` near the beginning of `main`, **before** any stochastic operation (including movement or multinomial sampling):

```
pal.Seed(1234)
```

The same seed reproduces the same random stream when the initial state and order of random calls also match. Stage 7 already seeds its complete program; earlier stages can add the same call.

To save the state at a chosen timestep, use Python's `pickle` to store the population grid, nutrient field, and current step number together. The following snippet belongs in ordinary Python code (for example, inside `main` after the simulation state exists), not inside an `@pal.njit` function:

```
import pickle

step = 0  # replace with the current timestep in your simulation
with open("cells.pkl", "wb") as f:
    pickle.dump({"cells": cells, "nutrient": nutrient, "step": step}, f)

with open("cells.pkl", "rb") as f:
    state = pickle.load(f)
cells = state["cells"]
nutrient = state["nutrient"]
step = state["step"]
```

Only load pickle files you trust: loading a malicious pickle can execute arbitrary Python code. Saving object state is not the same as saving every source of randomness or every external resource. To resume a stochastic trajectory exactly, also preserve the relevant random state and any other model state your experiment uses. A fixed seed is sufficient for replaying a run from its beginning under the same call sequence, not necessarily for resuming halfway through it.

Where to go next

You now have the central PAL workflow: construct state, visualize it, query neighborhoods, move cells, compile repeated work, evolve a nutrient field, switch cell representations while keeping that field, and record experiments. The remainder of this guide is organized by task. Start with Core use, Construction, and Shared spatial API for conventions common to the grids. For compilation and execution options, see Performance. Then consult the section for the component you need: Grid, AgentGrid, IList, PopGrid, PDEgrid, or Random numbers and Multinomial. The later sections cover saving PAL objects, organizing larger models, and visualization. You do not need to use every feature in the tutorial model.

Core use

This section collects import, execution-mode, and compilation conventions used throughout the component reference. The tutorial above works in PAL's default safe mode, so you can return to these options when you need them.

```
import PythonAutomataLibrary as pal
```

PAL starts in **safe mode**. Safe mode checks public operations and provides useful errors. To select fast mode:

```
pal.FastMode()
```

Call `FastMode()` before constructing any PAL model or window object. The first such construction locks the process-wide mode. Fast mode removes safety checks from performance-sensitive paths; it is intended for models already validated in safe mode.

Use `@pal.njit` for model computation:

```
@pal.njit
def Step(grid: pal.AgentGrid):
    ...
```

`pal.njit` wraps Numba `njit` and additionally transforms PAL model code. Among other PAL-specific compilation behavior, annotated PAL arguments allow safe-mode errors in compiled model code to report the model source line.

Construction

PAL uses `New...` functions to create its grids and helper objects. Choose the constructor for the kind of state you need; the table below summarizes the options before the individual component sections explain their operations.

Function	Purpose
`NewGrid(dimensions, dtype)`	Typed lattice values.
`NewAgentGrid(dimensions, numAgentProps=0, isStackable=False)`	Discrete or continuous agents with optional per-agent properties.
`NewPopGrid(dimensions, capacity=None)`	Integer population counts on a lattice.
`NewPDEgrid(dimensions)`	Continuous scalar field with diffusion/advection operations.
`NewIList()`	Reusable integer query/list container.
`NewMultinomial(other=None)`	Binomial/multinomial sampler.

Spatial grids use one to three dimensions. A **negative dimension enables wrapping on that axis**, while its absolute value gives the size. For example:

```
grid = pal.NewAgentGrid((100, 100))    # 100 x 100, no wrapping
grid = pal.NewAgentGrid((-100, 100))   # x wraps, y does not
grid = pal.NewPDEgrid((-100, -100))    # x and y wrap
```

`numAgentProps` controls how many floating-point properties each agent has. `isStackable=False` means at most one agent can occupy a lattice site; `True` allows multiple agents per site. `PopGrid.capacity` sets an optional **per-site** population limit, not a limit on the entire grid.

Constructor validation and limits

Each spatial dimension must be a nonzero integer within the supported int32 range (excluding the minimum signed value), and the product of absolute axis lengths must not exceed the int32 maximum. Fractional, nonfinite, zero, oversized, or malformed dimension inputs raise `ValueError` during construction. `AgentGrid` alone also accepts `()` for a zero-dimensional nonspatial population.

For validation, `numAgentProps` must be a nonnegative integer fitting AgentGrid's int32 metadata budget (which also reserves entries for spatial dimensions); fractional, nonfinite, negative, or oversized values raise `ValueError`. `isStackable` accepts booleans or `0` / `1`; other values raise `ValueError`.

`NewMultinomial(other)` accepts another PAL Multinomial in the **same safety mode** and copies its solver configuration, not its current sampling state. An unrelated object raises `TypeError`; mixing safe and fast samplers raises `ValueError`. Create a new sampler without `other` to use the default solver.

Shared spatial API

`Grid`, `AgentGrid`, `PopGrid`, and `PDEgrid` deliberately share spatial conventions. Geometry metadata (`xDim`, `yDim`, `zDim`, `nDims`, `wrapX`, `wrapY`, `wrapZ`) is read as attributes; geometry operations such as `ToI(...)` and `Box(...)` are methods.

Geometry

PAL identifies lattice sites either by their spatial coordinates, such as `(x, y)`, or by a single integer index. Coordinates are convenient for spatial rules; linear indices are useful for storing or sampling sites in integer collections. The conversion methods let you move between these representations.

Member	Meaning
`len(grid)`	Number of lattice sites.
`xDim`, `yDim`, `zDim`	Axis sizes. Missing axes are invalid in safe mode.
`nDims`	Number of spatial dimensions.
`wrapX`, `wrapY`, `wrapZ`	Whether each axis wraps.
`ToI(x, y=-1, z=-1)`	Convert coordinates to a linear lattice index.
`ItoX(i)`, `ItoY(i)`, `ItoZ(i)`	Recover coordinates from a linear index.
`InWrapX/Y/Z(value)`	Wrap an integer lattice coordinate on that axis, or return `-1` when outside a nonwrapped axis.

The same geometry names are used in 1D, 2D, and 3D. Supply only coordinates/axes that exist for the grid. `Grid`, `PopGrid`, and `PDEgrid` use `InWrapX/Y/Z` for lattice coordinates; `AgentGrid` distinguishes lattice `InWrapSQX/Y/Z` from continuous `InWrapX/Y/Z`.

Regions and neighborhoods

Use `Box` to visit sites in a rectangular region, and `Hood` to visit sites at specified offsets around a particular location. Both yield coordinates, so they can be used directly in spatial loops.

```
grid.Box(x1, x2)
grid.Box(x1, x2, y1, y2)
grid.Box(x1, x2, y1, y2, z1, z2)

grid.Hood(hood, x)
grid.Hood(hood, x, y)
grid.Hood(hood, x, y, z)
```

`Box` iterates coordinates in the half-open rectangular region `[x1, x2)`, `[y1, y2)`, `[z1, z2)` for the dimensions supplied. It yields an integer in 1D, `(x, y)` in 2D, and `(x, y, z)` in 3D. Bounds must be int32 integers, and the number of bound pairs must match grid dimensionality. On wrapped axes, out-of-range coordinates are wrapped; on nonwrapped axes, they are skipped. A box wider than a wrapped axis can therefore yield repeated sites. `Hood` maps relative offsets around a lattice position to the same coordinate forms, preserving offset order and duplicates; wrapped coordinates are mapped into the grid and nonwrapped out-of-range coordinates are omitted. Offset tuple lengths must match grid dimensionality. `Box` and `Hood` are direct-iteration constructs in compiled code: top-level `break` or `continue` from their loop body is unsupported; use explicit `range` loops when that control flow is needed.

Built-in neighborhoods:

```
pal.MooreHood(dim, excludeCenter=False)
pal.VonNeumannHood(dim, excludeCenter=False)
pal.CircleHood(dim, rad, excludeCenter=False)
```

`MooreHood` contains offsets with each coordinate in `[-1, 1]`. `VonNeumannHood` contains the center and axis-adjacent offsets. `CircleHood` contains integer offsets whose Euclidean distance from the center is at most `rad`; `rad` may be noninteger, but must be finite and nonnegative. Set `excludeCenter=True` to omit the zero offset. All three constructors support dimensions 1–3 and reject other dimension counts with `ValueError`. Each returns a tuple of integer offset tuples; these describe relative positions, not site indices, until mapped through `grid.Hood(...)`.

Performance: `FastMode()`, `njit(cache=True)`, and `unroll=True`

PAL provides three complementary performance controls:

- **`pal.FastMode()`** selects PAL's fast implementation, reducing safety-check overhead during execution. Call it once, before constructing any PAL model or window objects; otherwise PAL remains in safe mode. Fast mode is not a substitute for validating model correctness, and its speed advantage depends on the workload.
- **`@pal.njit(cache=True)`** compiles a Python function to native code, avoiding Python interpreter overhead in repeated numerical loops. The first call for a particular argument-type specialization still incurs compilation cost. `cache=True` saves compiled specializations to disk so later runs can reuse them when compatible; it does not make the first compilation faster, and some changes or environments require recompilation.
- **`unroll=True`** only affects performance inside functions compiled with `@pal.njit`; it provides no unrolling benefit in ordinary Python execution. It expands the loop body for each neighborhood offset, removing ordinary loop-control overhead and exposing individual operations to optimization. Use it only for **stencil operations over relatively small neighborhoods with**

simple loop bodies: both the number of neighbors and the amount of code inside the loop increase generated code size and compilation time. For large neighborhoods or complex loop bodies, prefer ordinary iteration.

These optimizations address different costs: **fast mode reduces runtime checks, JIT compilation reduces interpreter overhead, caching reduces repeated compilation across runs, and unrolling reduces inner-loop overhead**. They can be used together. Benchmark representative models because the benefit of each depends on the operations performed.

Example: unrolled Von Neumann neighborhood

For a small, fixed neighborhood, PAL can compile each neighbor operation directly into the function instead of executing a conventional loop. Consider summing values in the four cardinal directions around `(x, y)`:

```
VON_NEUMANN = pal.VonNeumannHood(2, excludeCenter=True)

@pal.njit(cache=True)
def NeighborSum(grid: pal.Grid, x: int, y: int):
    total = 0.0
    for nx, ny in grid.Hood(VON_NEUMANN, x, y, unroll=True):
        total += grid[nx, ny]
    return total
```

Conceptually, on an interior site of a nonwrapping grid, this is equivalent to:

```
total = 0.0
total += grid[x - 1, y]
total += grid[x + 1, y]
total += grid[x, y - 1]
total += grid[x, y + 1]
```

The exact offset order is not important for this sum; the example above illustrates the four operations, not the generated machine code. At boundaries, `Hood` still handles valid coordinates and wrapping according to the grid's configuration.

Why this is fast: The neighborhood has a fixed number of offsets known at compilation time. With `unroll=True`, Numba can specialize each operation, eliminating the ordinary neighborhood-loop control and making the resulting code easier to optimize. This can be **extremely performant** in the inner loop of an agent-based model, where the same small neighborhood is visited millions of times. It preserves PAL's compact neighborhood syntax rather than requiring manually expanded code.

The option changes how the loop is compiled, not which neighbors are visited. For large neighborhoods or substantial loop bodies, the duplicated operations can outweigh the runtime benefit; compare against ordinary iteration before enabling it.

Safety contracts and mode selection

Safe and fast objects expose the same modeling API. PAL starts in safe mode. To opt into fast mode, call `pal.FastMode()` **before constructing any PAL model or window object**. The first such construction locks the process-wide mode selection; a later `pal.FastMode()` raises `RuntimeError`. Run safe and fast comparisons in separate processes rather than trying to switch modes mid-run. Safe mode validates public operations and is the development/default mode; fast mode assumes the model obeys those contracts and removes performance-sensitive checks. Examples of safe-mode checks include lattice bounds and dimensionality, dead-agent and occupancy errors, PopGrid overflow, and PDE field/interface array shape or dimensionality. Invalid operations in fast mode are outside the supported contract rather than an alternative error-handling API.

Use safe mode until the model is correct, then benchmark fast mode. Do not rely on safe-mode exceptions as model control flow.

Grid

Use `Grid` when each lattice site needs a value, such as terrain type, occupancy flags, environmental measurements, or a precomputed spatial map. It is a general-purpose typed array on PAL's spatial lattice; unlike `AgentGrid`, it does not track individual agents, and unlike `PDEgrid`, it does not provide a transport solver.

```
import numpy as np

grid = pal.NewGrid((30, 30), np.float32)
grid[10, 12] = 2.5
```

The two dimensions specify the lattice size; the dtype specifies the type of value stored at each site. Use `np.float32` for a floating-point field, for example, or an integer dtype for categories or counts. Supported dtypes are `bool`, signed/unsigned 8/16/32/64-bit integers, and 32/64-bit floats.

`Grid` supports linear indexing, coordinate indexing, and NumPy-like slices:

```
grid[i]
grid[x, y]
grid[x, y, z]
grid[:] = 0
region = grid[2:5, 3:8]
```

Slice reads return detached NumPy copies. Scalar and slice reads/writes work in Python and compiled PAL model code. In safe mode, invalid scalar indices and coordinates are rejected; fast mode assumes valid access. Safe mode validates bounds, dimensionality, and whether assigned values are representable by the Grid dtype: boolean grids accept only `0` / `1`; integer grids reject fractional, nonfinite, and out-of-range assignments; floating grids reject nonfinite or out-of-range assignments. Use a wider dtype when a calculation may exceed its current range. A detached slice copy is not a live view of the Grid.

AgentGrid

Use `AgentGrid` when individual agents must retain their identities as they move, reproduce, die, or change properties. It manages agent handles and locations, making it suitable for cell-based and other individual-based models where tracking particular entities matters.

```
agents = pal.NewAgentGrid((xDim, yDim), numAgentProps=2)
```

PAL represents an agent by an integer handle. Agent properties are stored on the grid and accessed with `agents[agent, property]`.

Population and lifecycle

By default an `AgentGrid` is not stackable: at most one agent may occupy a lattice site. Set `isStackable=True` when multiple agents per site are required. Safe mode rejects creation or movement into an occupied site on a nonstackable grid without

changing model state.

Operation	Meaning
`GetPop()`	Number of living agents.
`Alive(agent)`	Whether an agent handle is alive.
`NewAgentSQ(...)`	Create an agent at a lattice site.
`NewAgent(...)`	Create an agent at a continuous position.
`Dispose(agent)`	Remove an agent.
`All(shuffle=False)`	Snapshot of all living agents; optionally shuffled.

`All()` has **snapshot semantics**. The returned collection can be iterated while agents are created or disposed without changing the current iteration. `All(shuffle=True)` returns the same population in randomized order.

```
for agent in agents.All():
    if ShouldDie(agent):
        agents.Dispose(agent)
```

Position

Discrete/lattice position:

```
agents.I(agent)
agents.XSQ(agent)
agents.YSQ(agent)
agents.ZSQ(agent)
agents.MoveSQ(agent, x, y, z)
```

Continuous position:

```
agents.X(agent)
agents.Y(agent)
agents.Z(agent)
agents.Move(agent, x, y, z)
```

`NewAgentSQ` and `MoveSQ` accept either a linear lattice index or lattice coordinates in every spatial dimension. SQ placement uses the center of the selected cell: after `NewAgentSQ(x, y)`, for example, `XSQ(agent)==x` and `YSQ(agent)==y`, while `X(agent)==x+0.5` and `Y(agent)==y+0.5`. Continuous `NewAgent` and `Move` instead keep the supplied spatial coordinates, with `XSQ` / `YSQ` / `ZSQ` identifying the containing cell.

Occupancy and queries

```
agents.LastAgent(x, y, z)
agents.AgentsAt(x, y, z)
```

```
agents.counts[x, y, z]
```

`LastAgent` returns the most recently stacked agent at a site, or `-1` if the site is empty. Iterate with `for agent in agents.AgentsAt(x, y):` in Python or inside `@pal.njit`. Do not create, dispose, or move agents during direct iteration, including moves within the same lattice site. Safe mode detects these changes and raises `RuntimeError`; fast mode omits that check. In Python, `AgentsAt(i)` also accepts a linear site index, including for multidimensional grids; `AgentsAt(x, y)` and `AgentsAt(x, y, z)` use coordinates. This overload is distinct from the continuous-position `AgentsInRadius` query. `counts` exposes lattice occupancy counts and supports linear-site, coordinate, and slice indexing (`agents.counts[i]`, `agents.counts[x, y]`, or `agents.counts[:, :]`). It is a **read-only query view**, not a second mutable population grid. Slice queries return detached arrays of occupancy counts. A nonspatial (`dimensions=()`) AgentGrid has no lattice counts; querying `len(agents.counts)` raises `ValueError`.

Radius iteration

`AgentsInRadius(rad, x[, y[, z]], exclude=None)` iterates agents near a continuous point in Python or inside `@pal.njit`. The yielded tuple depends on dimensionality:

```
for agent, dx in agents1.AgentsInRadius(rad, x):
    ...
for agent, dx, dy, distSq in agents2.AgentsInRadius(rad, x, y):
    ...
for agent, dx, dy, dz, distSq in agents3.AgentsInRadius(rad, x, y, z):
    ...
```

The radius is inclusive: agents with distance exactly `rad` are included. Returned displacements point from the query center to the agent and use the shortest periodic displacement on wrapped axes; `distSq` is their squared Euclidean distance in 2D/3D. `exclude` omits a specified agent handle. As with Python-side `AgentsAt`, structural modification during iteration is unsupported, and the generation check is enabled only in safe mode.

Wrapping

Discrete coordinates:

```
InWrapSQX(value)
InWrapSQY(value)
InWrapSQZ(value)
```

Continuous coordinates:

```
InWrapX(value)
InWrapY(value)
InWrapZ(value)
```

Wrapped displacement:

```
DispWrapX(x1, x2)
DispWrapY(y1, y2)
```

```
DispWrapZ(z1, z2)
```

`InWrapSQ*` maps integer lattice coordinates into the corresponding axis; `InWrap*` does the same for continuous coordinates. On a nonwrapped axis, an out-of-range coordinate returns `-1`. `DispWrap*(x1, x2)` returns the shortest signed displacement from `x1` to `x2`, accounting for periodic boundaries.

IList

Use `IList` to collect a reusable set of integer identifiers—such as grid indices or agent handles—for later iteration, shuffling, or random selection. It is useful for neighborhood queries, candidate destinations, and subsets of agents.

```
items = pal.NewIList()
```

`IList` collects integer identifiers for later use. These can be **grid-site indices** (such as `cells.ToI(x, y)`) or **agent handles**, letting you save a selection of sites or agents and subsequently iterate over it, shuffle it, or sample a random entry. For example, collect empty neighboring sites and choose one with `Random()`, or collect selected agents and process them later with `Iter()`. It stores identifiers, not copies of the sites or agents themselves. Stored values are nonnegative int32 integers in the supported contract.

Operation	Meaning
`Append(i)`	Append a nonnegative int32 value; returns the IList for chaining.
`Clear()`	Remove all entries; returns the IList for chaining.
`Random()`	Return a random entry.
`Shuffle()`	Shuffle entries in place; returns the IList for chaining.
`All()`	Return a detached copy.
`Iter()`	Return a no-copy/live iterable view.
`len(items)`, `items[i]`	Length/index access.

Use `Iter()` for one-pass iteration when a copy is unnecessary. Use `All()` when the returned values must remain independent of subsequent IList changes.

PopGrid

Use `PopGrid` when only the number of individuals at each lattice site matters, not their separate identities. It supports count-based population dynamics and movement, often avoiding the cost of representing every individual as a separate agent.

```
pop = pal.NewPopGrid((xDim, yDim), capacity=None)
```

`PopGrid` stores integer population counts. Indexing and direct assignment modify the current population immediately.

```
pop[x, y]
pop[x, y] = 10
pop[:] = 0
```

Linear (`pop[:]`) and spatial (`pop[:, :]`) slice reads return detached `int64` NumPy arrays; slice assignment writes to the population. Population changes can instead be accumulated and applied together:

```
pop.Add(delta, x, y)
pop.Update()
```

`Add` changes only the pending delta. Until `Update()` , `GetPop()` , indexing, and `All()` still reflect the current population. A single `Update()` commits the queued changes together, avoiding order-dependent reads during a synchronous timestep. Direct assignments during that timestep still change current values immediately.

Operation	Meaning
`GetPop()`	Total population over all sites.
`All()`	Copy of the linear indices of currently nonzero sites.
`Reset()`	Clear current populations and pending changes.

`capacity=None` uses the int64 maximum as the per-site population cap. An explicit `capacity` must be a nonnegative int64 integer; invalid, fractional, nonfinite, or out-of-range values raise `ValueError` during construction. Capacity constrains each site independently; total population is tracked separately and must remain representable as a nonnegative int64. In safe mode, direct writes enforce these bounds immediately. `Add()` queues signed changes without applying them to the current counts; a queued change may exceed a site's remaining capacity until it is checked at `Update()` . `Update()` validates every resulting site count and the total population before applying any pending delta; if validation fails, current populations and pending deltas are unchanged. Fast mode assumes these constraints are satisfied.

PDEgrid

Use `PDEgrid` for spatially varying continuous quantities that evolve through transport, such as nutrients, oxygen, signaling molecules, or temperature. It combines a scalar field with diffusion and advection operations, so it is preferable to a plain `Grid` when those dynamics are needed.

```
field = pal.NewPDEgrid((xDim, yDim))
```

`PDEgrid` stores a continuous `float32` scalar field. Direct indexing/assignment changes the current field immediately. Slice reads return detached `float32` NumPy arrays; array slice assignment writes to the field, but editing a previously read slice does not. Linear (`field[:]`) and spatial (`field[:, :]`) slices are supported. `Add(value, ...)` accumulates a pending delta; `Update()` applies pending changes simultaneously. `Reset()` clears both the field and pending changes.

Space and time

The time step `dt` specifies how much simulated time a transport update represents, while `dx`, `dy`, and `dz` specify the physical spacing between lattice sites along each axis. These values determine how diffusion and advection rates translate into changes on the grid.

```
field.SetTimeSpaceStep(dt, dx, dy=1.0, dz=1.0)
field.Dt()
field.Dx()
field.Dy()
field.Dz()
```

Transport methods use these spacings. `Dy()` and `Dz()` apply only when those dimensions exist; safe mode rejects missing-dimension access. `dt`, `dx`, `dy`, and `dz` must be finite and positive in safe mode.

Boundary conditions for transport

A boundary condition specifies what happens at the edge of the modeled region. PAL's Cartesian diffusion and advection methods use the same boundary argument names, but the physical effect depends on the transport process:

Boundary choice	How to specify it	Meaning
No flux	Omit the boundary argument, or pass `None`	No material crosses that nonwrapped face
Fixed exterior concentration	Pass a scalar, such as `xMinBC=1.0`	Prescribe the concentration just outside that face
Periodic	Construct the grid with a negative dimension on that axis, such as `(-30, 30)`	The two opposite faces connect; do not pass explicit BC values on that axis

The names identify both the **axis** (`x`, `y`, or `z`) and the **side** (`Min` or `Max`). For example, `xMinBC` controls the left face and `yMaxBC` the top face of a 2D grid. Unspecified nonwrapped faces are closed by default.

For example, on a nonwrapped 2D nutrient field:

```
field = pal.NewPDEgrid((30, 30))
field.SetTimeSpaceStep(1.0, 1.0, 1.0)
field.Diffusion(0.1, xMinBC=1.0, xMaxBC=0.0)
field.Update()
```

This prescribes an exterior concentration of 1 on the left and 0 on the right, with no flux across the top and bottom. The transport operation accumulates changes; `Update()` makes them current. Start with a nonuniform field or run multiple steps to see the resulting spatial profile.

For **diffusion**, the exterior concentration determines the gradient and therefore the boundary flux. For **advection**, it supplies the concentration entering through an inflow face; on an outflow face the upwind calculation uses the interior concentration instead. Thus the same `xMinBC` argument has a different numerical role in the two processes.

For a spatially varying boundary concentration, pass a C-contiguous `float32` array containing one value per site along the face. Scalar values are simpler when the whole face has the same condition. The boundary arguments apply to Cartesian methods;

radial diffusion instead uses `outerBC` for its outer radial face and symmetry at the center.

Diffusion

Diffusion spreads a quantity from regions of higher concentration toward regions of lower concentration. Use the diffusion methods to model processes such as nutrients or signaling molecules spreading through space; the variants below differ in how the diffusion rate is specified.

```
field.Diffusion(rateConstant, ...)
field.DiffusionMask(rateConstant, mask)
field.DiffusionField(rateConstants, ...)
field.DiffusionInterfaces(rateConstantsX, rateConstantsY=None, rateConstantsZ=None, ...)
field.DiffusionADI(rateConstant, ...)
field.DiffusionRadialCircle(rateConstant, outerBC=None)
field.DiffusionRadialSphere(rateConstant, outerBC=None)
```

`Diffusion` uses one constant diffusion rate. `DiffusionField` uses spatially varying cell-centered rates; each interior face uses the harmonic mean of its two adjacent cell rates, so a zero rate on either side blocks diffusion across that face. `DiffusionInterfaces` supplies the positive-face interface rate for each cell and axis directly; the negative face uses the neighboring cell's stored positive-face rate. `DiffusionMask` treats mask values `>= 0` as active and values `< 0` as blocked; diffusion occurs only between adjacent active cells, so the magnitude of a nonnegative mask value is irrelevant. `DiffusionADI` uses Crank-Nicolson in 1D, Peaceman-Rachford ADI in 2D, and Douglas-Gunn directional solves in 3D. `DiffusionRadialCircle` and `DiffusionRadialSphere` use radial shell finite-volume discretizations on a nonwrapped 1D grid with at least two cells. Cell `i` spans radii `i*dx` to `(i+1)*dx`, so cell 0 is the central disk/sphere and symmetry gives zero flux through its inner face. `outerBC` optionally sets the exterior concentration at the outer face.

Cartesian diffusion methods accept optional `xMinBC`, `xMaxBC`, `yMinBC`, `yMaxBC`, `zMinBC`, and `zMaxBC` fixed boundary values. A boundary may be a scalar applied uniformly across the face or a C-contiguous `float32` array with one value per site on that face; for example, an x-face array has `yDim*zDim` values in 3D. On a nonwrapped face, `None` means no flux through that face; wrapped axes are periodic and do not accept explicit boundary conditions. Field/interface arrays must match the grid geometry and dimensionality. Safe mode validates boundary size/type/finite values, explicit-scheme stability, and other inputs before changing pending state.

Advection

Advection transports a quantity through directed flow, such as a chemical carried by moving fluid. Unlike diffusion, which responds to concentration differences, advection follows a specified velocity field. The variants below differ in how velocity is represented.

```
field.Advection(vx, vy=0.0, vz=0.0, ...)
field.AdvectionField(xVels, yVels=None, zVels=None, ...)
field.AdvectionInterfaces(xVels, yVels=None, zVels=None, ...)
```

Advection uses a conservative first-order upwind discretization. `Advection` uses constant velocity components. `AdvectionField` uses cell-centered velocity fields and averages adjacent values at interior faces. `AdvectionInterfaces` supplies positive-face velocities directly; the negative face uses the neighboring cell's stored positive-face velocity. Supply velocity components only for dimensions that exist. The same scalar-or-face-array Cartesian boundary arguments used by diffusion are available: `None` closes a nonwrapped face to advective flux, while supplied values provide the exterior concentration used for inflow. Safe mode checks the CFL condition before accumulating changes.

Random numbers and Multinomial

Use PAL's random-number functions for stochastic decisions such as selecting agents or destinations. Use `Multinomial` when a count of individuals or events must be distributed among several possible outcomes without sampling each one separately.

```
pal.Seed(seed)
pal.Random()       # uniform float
pal.RandInt(max)  # integer in [0, max)
```

`Seed` accepts integral seeds in the uint64 domain (`0` through `2**64 - 1`); invalid or nonfinite inputs raise `ValueError`. `RandInt(max)` requires a positive integer in the int64 domain and returns an integer in `[0, max)`; zero, negative, fractional, nonfinite, or out-of-range bounds raise `ValueError`. `Random()` returns a uniform floating-point draw. PAL random functions share one seeded stream across Python and compiled PAL calls, so reseeding reproduces the same call sequence across that boundary. The stream is call-order dependent: adding or removing a random draw changes later results. This does not seed NumPy's separate RNG.

Multinomial allocations

A binomial draw counts how many of `n` trials have one outcome. A multinomial allocation extends this to several mutually exclusive outcomes, such as moving left, moving right, or staying put.

```
multi = pal.NewMultinomial()
x = multi.Binomial(100, 0.2)  # independent binomial draw

multi.Setup(100)
left = multi.Sample(0.2)
right = multi.Sample(0.3)
stay = 100 - left - right  # unallocated individuals
```

`Setup(n)` initializes a multinomial sequence with `n` trials and probability mass 1 and is chainable. Each `Sample(p)` takes that category's probability in the original multinomial distribution, draws from the remaining trials using the corresponding conditional probability, then removes the sampled count and `p` from the remaining state. Thus successive calls such as `Sample(0.2)` and `Sample(0.3)` represent categories with probabilities 0.2 and 0.3, not 0.2 followed by 0.3 of the remainder. In the example, the remaining probability is 0.5, and `stay` accounts for individuals not allocated to either sampled category. In safe mode cumulative requested probability cannot exceed 1. `Binomial(n, p)` performs an independent binomial draw and does not consume the current multinomial sequence. `NewMultinomial(other)` copies solver configuration but starts with fresh sampling state.

Reusing the binomial solver core

If several independent multinomial allocations are needed, construct one sampler and pass it into another constructor:

```
movement = pal.NewMultinomial()
division = pal.NewMultinomial(movement)

movement.Setup(100)
moved = movement.Sample(0.2)

division.Setup(50)
dividing = division.Sample(0.1)
```

`NewMultinomial(movement)` **shares the underlying reusable binomial solver core**, avoiding a separate solver setup, but the two `Multinomial` objects keep **independent remaining-count and remaining-probability state**. The second instance does not inherit the first instance's partially sampled `Setup` sequence. Call `Setup` separately for each allocation. Sharing the solver does not create separate random streams: both samplers use PAL's common RNG, so results still depend on call order.

Saving PAL objects

Use Python's standard `pickle` module to save PAL model objects and reload them later. This is useful for checkpoints and reproducible experiments; tutorial stage 8 shows a complete save-and-reload example. Only load pickle files from sources you trust.

What a checkpoint preserves

PAL saves model state rather than process-local pointers. The saved contents depend on the object:

Object	Preserved state
`AgentGrid`	Geometry, occupancy, agents, and properties
`PopGrid`	Geometry, capacity, current counts, and pending deltas
`PDEgrid`	Geometry, time/space steps, current field, and pending deltas
`IList`	Stored integer identifiers
`Multinomial`	Current `Setup` / `Sample` remainder

Safe/fast identity is also restored. In particular, a `PopGrid` or `PDEgrid` can be saved before `Update()` without losing its queued changes.

Organizing larger models

Once a model uses several grids and parameter arrays, it can be useful to group them into a single object. This is an optional organizational technique, not a requirement for using PAL.

Package model components with a named tuple

As models grow, passing every grid and parameter separately can make function signatures difficult to read. Python's `typing.NamedTuple` groups related objects under descriptive field names, and Numba supports accessing those fields inside compiled functions.

For example, package an agent grid, a nutrient field, and a NumPy array of rates. The three named-tuple fields hold references to those objects; the grids and array allocate their own storage. The named tuple does not need to reserve space for every cell or nutrient value:

```
from typing import NamedTuple
import numpy as np
import PythonAutomataLibrary as pal

class Model(NamedTuple):
```

```
    cells: pal.AgentGrid
    nutrient: pal.PDEgrid
    rates: np.ndarray

@pal.njit(cache=True)
def Consume(model: Model):
    for agent in model.cells.All():
        x, y = model.cells.XSQ(agent), model.cells.YSQ(agent)
        model.nutrient[x, y] -= model.rates[0]

@pal.njit(cache=True)
def Step(model: Model):
    Consume(model)
    model.rates[1] += 1.0

def main():
    cells = pal.NewAgentGrid((8, 8), isStackable=False)
    cells.NewAgentSQ(3, 3)
    nutrient = pal.NewPDEgrid((8, 8))
    nutrient[:] = 1.0
    model = Model(cells=cells, nutrient=nutrient, rates=np.array([0.1, 0.0]))
    Step(model)
    print(model.nutrient[3, 3], model.rates[1])

if __name__ == "__main__":
    main()
```

The `NamedTuple` class gives editors field names and type information for autocomplete, including `model.cells` and `model.nutrient`. Both `Step` and `Consume` receive the same named model, and changes to the contained grids and arrays persist. The named tuple itself is immutable: `model.cells = other_cells` is not allowed. For a changing scalar, use a mutable array as shown with `rates[1]`, or return an updated value.

Performance: Numba compiles named-field access, but passing a composite from Python into a compiled function can have conversion overhead. Prefer calling multiple compiled functions from a compiled timestep, as above, rather than repeatedly entering compiled code for tiny operations. Whether the composite is faster than separate arguments depends on the workload; benchmark performance-sensitive applications. See Numba's supported Python features.

Visualization

Use PAL's visualization components to inspect spatial model state while a simulation runs. `PixWindow` is suited to pixel-based lattice views, while the OpenGL windows support 2D and 3D geometry; rendering and simulation computation remain separate.

PixWindow

Choose `PixWindow` for a two-dimensional lattice view where each site or pixel is assigned a color. It is well suited to displaying cell occupancy, population counts, and scalar fields without drawing individual geometric shapes.

```
pix, window = pal.StartPixWindow(xDim, yDim, scale=1, title="PAL", headless=False)
```

`pix` is a write-oriented pixel buffer, not a general-purpose numeric `Grid`: it supports assignment but does not expose a public pixel-read API. Assign integer RGB colors by coordinate or linear index:

```
pix[x, y] = 0xFF0000
pix[i] = 0x00FF00
```

It also provides `xDim`, `yDim`, `ToI(x, y)`, `ItoX(i)`, `ItoY(i)`, and `len(pix)`.

The window lifecycle is:

```
window.Update()
window.Save(path, block=False)
window.StartGif(path, delay=100)
window.AddGifFrame(block=False)
window.StopGif()
window.IsOpen()
window.Close()
```

`Update()` publishes the current pixel buffer to the renderer. `StartGif()` begins a GIF recording, each `AddGifFrame()` appends the currently rendered frame, and `StopGif()` finalizes the file. `block=True` on asynchronous output operations waits for that requested output to complete before returning; call `StopGif()` before relying on the GIF file as complete.

OpenGLWindow

Choose `OpenGLWindow` when you want to draw geometric objects such as circles, boxes, and lines, including in three dimensions. Unlike `PixWindow`, it renders shapes rather than requiring you to color individual pixels.

```
draw, window = pal.StartOpenGLWindow(xDim, yDim, zDim=None,
                                     width=800, height=800,
                                     title="PAL", headless=False)
```

Draw geometry with:

```
draw.Circle(rad, color, x, y, z=0.0)
draw.Box(xLen, color, x, y, z=0.0, yLen=None, zLen=None)
draw.BoxSQ(color, x, y, z=0.0)
draw.Line(width, color, x1, y1, x2, y2, z1=0.0, z2=0.0)
```

Window controls include `Borders(width, color)`, `Camera(x, y, z, yaw=None, pitch=None)`, `Background(color)`, `Clear()`, `Update()`, `IsOpen()`, `Save(path, block=False)`, `StartGif(path, delay=100)`, `AddGifFrame(block=False, timeout=30)`, `StopGif(timeout=30)`, and `Close()`. The GIF lifecycle is the same as for `PixWindow`: start recording, append rendered frames, then stop to finalize the file.

Colors are packed integer RGB values such as `0xFF0000`. `pal.ColorScale(colors, value)` interpolates RGB channels across a sequence of colors for normalized `value`; values outside `[0, 1]` clamp to the endpoint colors.

Set `headless=True` for off-screen image/GIF generation. OpenGL headless rendering requires a supported standalone backend such as EGL.

Use `pal.AwaitWindows()` when model execution should wait for PAL windows to finish their lifecycle.